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Exercise # 14.3

Q.4: $\int_0^{\pi/6} \int_0^{\cos 3\theta} r\,dr\,d\theta$

$$= \int_0^{\pi/6} \left[\int_0^{\cos 3\theta} r\,dr \right] d\theta$$

$$\int_0^{\cos 3\theta} r\,dr = \left[\frac{r^2}{2}\right]_0^{\cos 3\theta}$$

evaluating the limits in the variable r

$$\frac{(\cos 3\theta)^2}{2} - \frac{(0)^2}{2} = \frac{\cos^2 3\theta}{2}$$

Simplify by using trigonometeric functions or identities.

$$\frac{1}{2}\left(\frac{1+\cos\theta}{2}\right) = \frac{1+\cos\theta}{4}$$

$$\int_0^{\pi/6}\left[\int_0^{\cos 3\theta} r\,dr\right] d\theta = \int_0^{\pi/6}\left(\frac{1+\cos 6\theta}{4}\right) d\theta$$

$$= \frac{1}{4}\int_0^{\pi/6} (1+\cos 6\theta)\,d\theta$$

integrating $\frac{1}{4}\left(\theta + \frac{1}{6}\sin 6\theta\right)_0^{\pi/6}$

$$\frac{1}{4}\left(\frac{\pi}{6} + \frac{1}{6}\sin 6\left(\frac{\pi}{6}\right)\right) - \frac{1}{4}\left(0 + \frac{1}{6}\sin 6(0)\right)$$

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$$= \int_0^{\pi/2} \left[\left(-x\cos\frac{\pi}{3} - \frac{\sin x}{2}\left(\frac{\pi}{3}\right)^2 \right] - \left[-x\cos 0 - \frac{\sin x(0)^2}{2} \right] dx$$

$$= \int_0^{\pi/2} \left[-\frac{x}{2} - \frac{\pi^2}{18}\sin x + x \right] dx = \int_0^{\pi/2} \left(\frac{x}{2} - \frac{\pi^2}{18}\sin x \right) dx$$

$$= \left| \frac{x^2}{4} - \frac{\pi^2}{18}(-\cos x) \right|_0^{\pi/2}$$

$$= \left(\frac{\left(\frac{\pi}{2}\right)^2}{4} - \frac{\pi^2}{18}\cos\frac{\pi}{2} \right) - \left(\frac{(0)^2}{4} + \frac{\pi^2}{18}\cos 0 \right)$$

$$= \frac{\pi^2}{16} + 0 - 0 - \frac{\pi^2}{18} = \frac{9\pi^2 - 8\pi^2}{144} = \boxed{\frac{\pi^2}{144}}$$ Ans

Exercise # 14.2

Q.6: $\int_{-1}^{1}\int_{-x^2}^{x^2}(x^2-y)\,dy\,dx = \int_{-1}^{1}\int$

$$= \int_{-1}^{1} \left[\int_{-x^2}^{x^2} (x^2-y)\,dy \right] dx$$

$$= \int_{-1}^{1} \left[x^2y - \frac{y^2}{2} \right]_{-x^2}^{x^2} dx$$

$$= \int_{-1}^{1} \left(\left(x^2(x^2) - \frac{(x^2)^2}{2} \right) - \left(x^2(-x^2) - \left(\frac{-x^2}{2}\right)^2 \right] dx$$

$$= \int_{-1}^{1} 2x^4\,dx = \left[\frac{2x^5}{5} \right]_{-1}^{1}$$

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$\frac{2}{5}\left[(1)^5-(-1)^5\right] = \frac{2}{5}[1+1] = \boxed{\frac{4}{5}}$ Ans

Q.7: $\int_0^1 \int_0^x y\sqrt{x^2-y^2}\,dy\,dx$

$= \int_0^1 \left[\int_0^x y\sqrt{x^2-y^2}\,dy\right]dx$

$= \int_0^1 \left[\frac{-1}{2}\times\frac{2}{3}(x^2-y^2)^{3/2}\right]dx$

$= \int_0^1 \left[-\frac{1}{3}\left\{(x^2-(x)^2)^{3/2} - (x^2-(0)^2)^{3/2}\right]dx$

$= \int_0^1 \frac{1}{3}x^3\,dx = \frac{1}{3}\left[\frac{x^4}{4}\right]_0^1 = \left[\frac{x^4}{12}\right]_0^1$

$= \frac{1}{12}\left[(1)^4-(0)^4\right] = \boxed{\frac{1}{12}}$ Ans

Q.8: $\int_1^2 \int_0^{y^2} e^{x/y^2}\,dx\,dy$

$= \int_1^2 \left\{y^2\int_0^{y^2} e^{x/y^2}\cdot\frac{1}{y^2}\,dx = \int_1^2 y^2\left(e^{x/y^2}\right)_0^{y^2}\right.$

$= \int_1^2 \left\{y^2\left[e^{y^2/y^2} - e^{0/y^2}\right]dy\right.$

$= \int_1^2 \left\{y^2\left[e^1-e^0\right]dy\right.$

$= \int_1^2 (e-1)y^2\,dy = (e-1)\left[\frac{y^3}{3}\right]_1^2$

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$$= \frac{(e-1)}{3}\left[(2)^3-(1)^3\right] = \boxed{\frac{7(e-1)}{3}}$$ Ans

Q.19 $\iint_R x(1+y^2)^{-1/2}\,dA$; R is the region in the first quadrant enclosed by $y=x^2$, $y=4$ and $x=0$

$$= \int_0^4 \int_0^{\sqrt{y}} x(1+y^2)^{-1/2}\,dx\,dy$$

$$= \int_0^4 \frac{1}{2}y(1+y^2)^{-1/2}\,dy = \boxed{\frac{\sqrt{17}-1}{2}}$$ Ans

Q.56: $\iint_D x\,dA = \int_1^3 \int_0^{\ln x} x\,dy\,dx$

$$\iint_D x\,dA = \int_0^{\ln 3} \int_{e^y}^{3} x\,dx\,dy$$

$$= \int_0^{\ln 3}\left[\frac{x^2}{2}\right]_{x=e^y}^{x=3} dy = \frac{1}{2}\int_0^{\ln 3} 9-e^{2y}\,dy$$

$$= \frac{1}{2}\left[-\frac{1}{2}e^{2y}+9y\right]_0^{\ln 3}$$

$$= \frac{1}{2}\left[-\frac{1}{2}e^{2\ln 3}+9\ln 3\right]-\frac{1}{2}\left[0-\frac{1}{2}\right]$$

$$= \frac{1}{2}\left[9\ln 3-\frac{9}{2}+\frac{1}{2}\right] = \boxed{-2+\frac{9}{2}\ln 3}$$ Ans

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$\frac{1}{4}\left(\frac{\pi}{6}+0\right)-\frac{1}{4}(0)=\boxed{\frac{\pi}{24}}$ Ans

Q.5: $\int_0^{\pi}\int_0^{1-\sin\theta} r^2\cos\theta\, dr\, d\theta$

$=\int_0^{\pi}\int_0^{1-\sin\theta} r^2\cos\theta\, dr\, d\theta$

$=\int_0^{\pi}\left[\int_0^{1-\sin\theta} r^2\cos\theta\, dr\right]d\theta$

Solve the inner integral and treat θ as a constant.

$=\cos\theta\int_0^{1-\sin\theta} r^2\, dr = \cos\theta\left[\frac{r^3}{3}\right]_0^{1-\sin\theta}$

evalulating the limits in the variable r

$\cos\theta\left[\frac{(1-\sin\theta)^3}{3}-\frac{(0)^3}{3}\right]=\frac{(1-\sin\theta)^3}{3}\cos\theta$

$\int_0^{\pi}\left[\int_0^{\cos 3\theta} r\,dr\right]d\theta=\int_0^{\pi}\frac{(1-\sin\theta)^3\cos\theta\, d\theta}{3}$

$\int_0^{\pi}\left[\int_0^{\pi}(1-\sin\theta)^3\cos\theta\, d\theta\right.$

$=\frac{-1}{3}\int_0^{\pi}(1-\sin\theta)^3(-\cos\theta)\,d\theta$

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integrating $\frac{-1}{3}\left[\frac{(1-\sin\theta)^4}{4}\right]_0^{\pi}$

$= \frac{1}{12}\left((1-\sin\theta)^4\right)_{\pi}^{0}$

evalutate $\frac{1}{12}\left((1-\sin 0)^4 - (1-\sin\pi)^4\right]$

$= \frac{1}{12}\left((1)^4 - (1)^4\right) = \boxed{0}$ Ans

Q.6: $\int_0^{\pi/2}\int_0^{\cos\theta} r^3\,dr\,d\theta$

$\rightarrow \int_0^{\pi/2}\left[\int_0^{\cos\theta} r^3\,dr\right]d\theta$

Solve the inner integral and treat θ as a constant.

$= \int_0^{\cos\theta} r^3\,dr = \left[\frac{r^4}{4}\right]_0^{\cos\theta}$

$= \frac{1}{4}\cos^4\theta$

$\rightarrow \int_0^{\pi/2}\left[\int_0^{\cos\theta} r^3\,dr\right]d\theta = \int_0^{\pi/2}\frac{1}{4}\cos^4\theta\,d\theta$

$= \frac{1}{4}\int_0^{\pi/2}\cos^4\theta\,d\theta$

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use the identity $\cos^2 x = \dfrac{1+\cos 2x}{2}$

$$= \frac{1}{4}\int_0^{\pi/2}\left(\frac{1+\cos 2\theta}{2}\right)d\theta$$

$$= \frac{1}{4}\int_0^{\pi/2}\left(\frac{1+2\cos 2\theta+\cos^2 2\theta}{2}\right)d\theta$$

$$= \frac{1}{16}\int_0^{\pi/2}\left(1+2\cos 2\theta+\frac{1}{2}+\frac{\cos 4\theta}{2}\right)d\theta$$

$$= \frac{1}{16}\int_0^{\pi/2}\left(\frac{3}{2}+2\cos 2\theta+\frac{\cos 4\theta}{2}\right)d\theta$$

integrating $\dfrac{1}{16}\left(\dfrac{3}{2}\theta+\sin 2\theta+\dfrac{1}{8}\sin 4\theta\right)\Big|_0^{\pi/2}$

evalulate,

$$= \frac{1}{16}\left(\frac{3}{2}\left(\frac{\pi}{2}\right)+\sin 2\left(\frac{\pi}{2}\right)+\frac{1}{8}\sin 4\left(\frac{\pi}{2}\right)\right)$$

$$= -\frac{1}{16}\left(\frac{3}{2}(0)+\sin 2(0)+\frac{1}{8}\sin 4(0)\right)$$

$$= \frac{1}{16}\left(\frac{3}{2}\left(\frac{\pi}{2}\right)+0\right)-\frac{1}{16}(0) = \boxed{\frac{3}{64}\pi}$$ Ans

Q.7: The region enclosed by the cardioid $r = 1-\cos\theta$.

$$r = -\cos\theta+1,\ 0 \le \theta \le 2\pi$$

$$\int_0^{2\pi}\int_0^{1-\cos\theta} r\,dr\,d\theta = A$$

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Evaluate,

$$A = \frac{1}{2}\int_0^{2\pi} (-\cos\theta + 1)^2 \, d\theta$$

$$A = \frac{1}{2}\int_0^{2\pi} \left(1 - 2\cos\theta + \cos^2\theta + \frac{\cos 2\theta}{2}\right) d\theta$$

$$A = \frac{1}{2}\int_0^{2\pi} \left(1 - 2\cos\theta + \frac{1}{2} + \frac{\cos 2\theta}{2}\right) d\theta$$

$$A = \frac{1}{2}\int_0^{2\pi} \left(\frac{3}{2} - 2\cos\theta + \frac{\cos 2\theta}{2}\right) d\theta$$

integrate,

$$\frac{1}{2}\left[-2\sin\theta + \frac{\sin 2\theta}{4} + \frac{3}{2}\theta\right]_0^{2\pi} = A$$

$$\left[\frac{6\pi}{2} - 2\sin 2\pi + \frac{\sin 4\pi}{4}\right]\frac{1}{2} - \frac{1}{2}[0] = A$$

$$\left(\frac{6\pi}{6}\right)\frac{1}{2} = A \qquad \boxed{\frac{3\pi}{2} = A}$$

Q.8: The region enclosed by the rose $r = \sin 2\theta$

$\sin 2\theta = r, \; 0 \le \theta \le \pi/2$

So $A = \int_0^{\pi/2}\int_0^{\sin\theta} r \, dr \, d\theta$

integrate with respect to r

$$4\int_0^{\pi/2}\left[\frac{r^2}{2}\right]_0^{\sin\theta} d\theta = A$$

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$$\int_0^{\pi/2} \left[r^2\right]_0^{\sin 2\pi} d\theta = A$$

Evaluate

$$A = 2\int_0^{\pi/2} \sin^2 2\theta \, d\theta$$

Evaluate

$$2\int_0^{\pi/2} \sin^2 2\theta \, d\theta - A$$

$$2\int_0^{\pi/2} \left[\frac{1}{2} - \frac{\cos 4\theta}{2}\right]_0^2 d\theta = A$$

integrate

$$A = \left[-\frac{\sin 4\theta}{4} + \theta\right]_0^{\pi/2}$$

$$A = \left[-\frac{\sin 2\pi}{4} + \frac{\pi}{2}\right] - [0]$$

$$\boxed{A = \frac{\pi}{2}}$$